



GSFA Online

Updated up to the 48th Session of the Codex Alimentarius Commission (2025)

FOOD ADDITIVE DETAILS

Citric and fatty acid esters of glycerol (472c)

Synonym(s)

- CITREM
- Citric acid esters of mono- and di-glycerides
- Citroglycerides

Functional Classes

- Antioxidant
- Emulsifier
- Flour treatment agent
- Sequestrant
- Stabilizer

[Click here to search the FAO JECFA database for the specifications of additive\(s\) with INS No. 472c](#)

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CSV (Tables)

GSFA Provisions for Citric and fatty acid esters of glycerol

Number	Food Category	Max Level	Notes
14.1.5	Coffee, coffee substitutes, tea, herbal infusions, and other hot cereal and grain beverages, excluding cocoa		GMP Note 160
13.2	Complementary foods for infants and young children	5,000 mg/kg	Note XS73 Note 268
09.2.4.1	Cooked fish and fish products		GMP Note 241
01.2.1.2	Fermented milks (plain), heat-treated after fermentation		GMP Note 234
01.2.1.1	Fermented milks (plain), not heat-treated after fermentation		GMP Note 234 Note 235 Note 634
04.2.2.7	Fermented vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera) and seaweed products, excluding fermented soybean products of food categories 06.8.6, 06.8.7, 12.9.1, 12.9.2.1 and 12.9.2.3		GMP Note XS294
13.1.3	Formulae for special medical purposes for infants	9,000 mg/kg	Note 381 Note 551 Note 380
08.1.2	Fresh meat, poultry, and game, comminuted		GMP Note 281
08.1.1	Fresh meat, poultry, and game, whole pieces or cuts		GMP Note 16 Note 326
09.1.2	Fresh mollusks, crustaceans, and echinoderms		GMP Note XS315 Note 390 Note XS312
06.4.1	Fresh pastas and noodles and like products		GMP Note 211
09.2.4.3	Fried fish and fish products, including mollusks, crustaceans, and echinoderms		GMP Note 41
09.2.2	Frozen battered fish, fish fillets, and fish products, including mollusks, crustaceans, and echinoderms		GMP Note 16 Note XS166

Number	Food Category	Max Level	Notes
 09.2.1	Frozen fish, fish fillets, and fish products, including mollusks, crustaceans, and echinoderms	GMP	 Note XS95  Note XS312  Note XS36  Note XS191  Note XS315  Note XS92  Note 392  Note XS165  Note XS190
 09.2.3	Frozen minced and creamed fish products, including mollusks, crustaceans, and echinoderms	GMP	 Note 16
 13.1.1	Infant formulae	9,000 mg/kg	 Note 381  Note 551  Note 380
 02.1.3	Lard, tallow, fish oil, and other animal fats	100 mg/kg	 Note 521
 01.1.2	Other fluid milk (plain)	GMP	 Note 407
 11.4	Other sugars and syrups (e.g. xylose, maple syrup, sugar toppings)	GMP	 Note 258
 01.4.1	Pasteurized cream (plain)	GMP	 Note XS288
 01.2.2	Renneted milk (plain)	GMP	
 12.1.2	Salt Substitutes	GMP	
 09.2.5	Smoked, dried, fermented, and/or salted fish and fish products, including mollusks, crustaceans, and echinoderms	GMP	 Note XS236  Note XS167  Note 300  Note XS222  Note XS311  Note XS189  Note XS244
 01.4.2	Sterilized and UHT creams, whipping and whipped creams, and reduced fat creams (plain)	GMP	
 02.1.2	Vegetable oils and fats	100 mg/kg	 Note XS33  Note 520  Note XS325R  Note 511

Note: Unless otherwise specified, food additive provisions apply to the food category indicated (e.g. Dairy), as well as to all subcategories of that category (e.g. Cheese, Ripened Cheese, etc.).

GSFA Table 3 Provisions

Citric and fatty acid esters of glycerol is a food additive that is included in **Table 3**, and as such may be used in the following foods under the conditions of good manufacturing practices (GMP) as outlined in the Preamble of the Codex GSFA. Although not listed below, Citric and fatty acid esters of glycerol could also be used in heat-treated butter milk of food category 01.1.1 and spices of food category 12.2.1. Note that food categories listed in the **Annex to Table 3** were excluded accordingly. **Citric and fatty acid esters of glycerol** is acceptable in foods conforming to the following commodity standards: CS 275-1973, CS 253-2006 (see functional class table and footnote), CG 95-2022 (For use at 9000 mg/kg as emulsifier).

Citric and fatty acid esters of glycerol is a: Antioxidant, Emulsifier, Stabilizer, Thickener

Any Antioxidant listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981, CS 291-2010, CS 319-2015 (canned mangoes only), CS 256-2007, CS 306-2011.

Any Emulsifier listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981, CS 309R-2011, CS 256-2007.

Any Stabilizer listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981, CS 256-2007.

Any Thickener listed in Table 3 can be acceptable for use in all products conforming to CS 117-1981.

Number	Food Category
 01.1.4	Flavoured fluid milk drinks
 01.3	Condensed milk and analogues (plain)
 01.4.3	Clotted cream (plain)
 01.4.4	Cream analogues
 01.5	Milk powder and cream powder and powder analogues (plain)
 01.6.1	Unripened cheese
 01.6.2	Ripened cheese

Number ▼	Food Category ▼
 01.6.4	Processed cheese
 01.6.5	Cheese analogues
 01.7	Dairy-based desserts (e.g. pudding, fruit or flavoured yoghurt)
 01.8.1	Liquid whey and whey products, excluding whey cheeses
 02.2.2	Fat spreads, dairy fat spreads and blended spreads
 02.3	Fat emulsions mainly of type oil-in-water, including mixed and/or flavoured products based on fat emulsions
 02.4	Fat-based desserts excluding dairy-based dessert products of food category 01.7
 03.0	Edible ices, including sherbet and sorbet
 04.1.2	Processed fruit
 04.2.2.2	Dried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweeds, and nuts and seeds
 04.2.2.3	Vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds in vinegar, oil, brine, or soybean sauce
 04.2.2.4	Canned or bottled (pasteurized) or retort pouch vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds
 04.2.2.5	Vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed purees and spreads (e.g., peanut butter)
 04.2.2.6	Vegetable (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), seaweed, and nut and seed pulps and preparations (e.g. vegetable desserts and sauces, candied vegetables) other than food category 04.2.2.5
 04.2.2.8	Cooked or fried vegetables (including mushrooms and fungi, roots and tubers, pulses and legumes, and aloe vera), and seaweeds
 05.0	Confectionery
 06.3	Breakfast cereals, including rolled oats
 06.4.3	Pre-cooked pastas and noodles and like products
 06.5	Cereal and starch based desserts (e.g. rice pudding, tapioca pudding)
 06.6	Batters (e.g. for breading or batters for fish or poultry)
 06.7	Pre-cooked or processed rice products, including rice cakes (Oriental type only)
 06.8	Soybean products (excluding soybean-based seasonings and condiments of food category 12.9)
 07.0	Bakery wares
 08.2	Processed meat, poultry, and game products in whole pieces or cuts
 08.3	Processed comminuted meat, poultry, and game products
 08.4	Edible casings (e.g. sausage casings)
 09.3	Semi-preserved fish and fish products, including mollusks, crustaceans, and echinoderms
 09.4	Fully preserved, including canned or fermented fish and fish products, including mollusks, crustaceans, and echinoderms
 10.2.3	Dried and/or heat coagulated egg products
 10.3	Preserved eggs, including alkaline, salted, and canned eggs
 10.4	Egg-based desserts (e.g. custard)
 11.6	Table-top sweeteners, including those containing high-intensity sweeteners
 12.2.2	Seasonings and condiments
 12.3	Vinegars
 12.4	Mustards
 12.5	Soups and broths
 12.6	Sauces and like products
 12.7	Salads (e.g. macaroni salad, potato salad) and sandwich spreads excluding cocoa- and nut-based spreads of food categories 04.2.2.5 and 05.1.3
 12.8	Yeast and like products
 12.9	Soybean-based seasonings and condiments
 12.10	Protein products other than from soybeans
 13.3	Dietetic foods intended for special medical purposes (excluding products of food category 13.1)

Number ▾	Food Category ▾
 13.4	Dietetic formulae for slimming purposes and weight reduction
 13.5	Dietetic foods (e.g. supplementary foods for dietary use) excluding products of food categories 13.1 - 13.4 and 13.6
 13.6	Food supplements
 14.1.4	Water-based flavoured drinks, including "sport," "energy," or "electrolyte" drinks and particulated drinks
 14.2.1	Beer and malt beverages
 14.2.2	Cider and perry
 14.2.4	Wines (other than grape)
 14.2.5	Mead
 14.2.6	Distilled spirituous beverages containing more than 15% alcohol
 14.2.7	Aromatized alcoholic beverages (e.g. beer, wine and spirituous cooler-type beverages, low alcoholic refreshers)
 15.0	Ready-to-eat savouries
 16.0	Prepared foods

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